

THE NATIONAL INSTITUTE OF ENGINEERING, MYSURU

(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)



REPORT ON

Project Title: Device Anomaly Detection

Project Description: Dynamic correlation of device failure logs and development of an algorithm to determine failure patterns

Submitted By:

Name: Samanth B Upadhye

USN: 4NI23CI092

Branch: AIML 'B'

Semester: '5th'

Submitted To:

Balaji V

Assistant Professor

Department of CS&E

NIE, Mysuru

Date of Submission: 30-10-2025

TABLE OF CONTENTS

- Abstract
- Introduction
- Problem Statement
- Objectives
- Literature Survey
- Proposed Methodology
- Expected Outcomes
- Challenges And Limitations
- Future Scope
- Conclusion
- References

ABSTRACT

In modern computing environments, the early detection of hardware and firmware anomalies is crucial to prevent unexpected device failures, system downtime, and user dissatisfaction. With the growing interconnectivity of devices and the rapid increase in data complexity, manual monitoring of logs and error events has become inefficient and error-prone.

This project focuses on dynamically correlating device failure logs and developing an intelligent algorithm capable of identifying and predicting failure patterns before they evolve into critical faults. By analysing large-scale log data, the proposed system aims to learn normal operational behaviour, detect deviations, and uncover hidden relationships among failure-inducing parameters.

Through the application of advanced machine learning techniques such as Isolation Forests and Autoencoders, the project seeks to create a data-driven framework that enhances the reliability, maintainability, and performance of computing devices. The ultimate goal is to enable proactive fault detection and support predictive maintenance strategies for improved device longevity.

INTRODUCTION

- **Background And Motivation**

As computing devices become more complex, ensuring their reliability is increasingly critical. Every system - whether a laptop, server, or IoT device - generates large volumes of operational logs that hold valuable insights into performance and health. However, the scale and unstructured nature of these logs make manual analysis impractical.

This project aims to apply machine learning to automatically identify hidden failure patterns and early warning signs that traditional rule-based monitoring often misses.

- **Why Device Anomaly Detection Is Important?**

Static threshold-based monitoring cannot capture subtle deviations in device behaviour. In contrast, anomaly detection enables adaptive learning from real-time data, allowing proactive fault prediction. This approach reduces downtime, lowers maintenance costs, and improves user experience.

- **Real-World Implications**

Industries like Intel and Dell already use predictive maintenance systems to forecast hardware issues and optimize repairs. Inspired by these, this project seeks to develop an intelligent, self-monitoring framework that enhances reliability and system resilience.

PROBLEM STATEMENT

Modern computing devices generate large volumes of system and failure logs that capture valuable insights about performance, health, and potential degradation. However, these logs are often unstructured, heterogeneous, and produced at high frequency, making manual monitoring inefficient and error-prone.

This project aims to address the problem of dynamic correlation of device failure logs to uncover meaningful relationships between various system parameters and their connection to device malfunctions. *Dynamic correlation* involves analysing temporal and contextual relationships among multiple log sources to detect evolving patterns that precede failure events.

The goal is to build a machine learning-based anomaly detection algorithm that can learn normal device behaviour and automatically identify deviations signalling potential failures. This approach will enable proactive fault detection and predictive maintenance, reduce unplanned downtime and improve overall device reliability.

OBJECTIVES

The primary objective of this project is to develop a machine learning-based framework that detects anomalies and predicts potential device failures through intelligent correlation of system and failure logs. The aim is to move from manual, rule-based inspection to an automated, data-driven approach capable of recognizing early signs of malfunction.

The specific objectives of the project are as follows:

- 1. Log Data Understanding and Preprocessing:** Analyse system logs, clean and normalize data, and extract key features for analysis.
- 2. Dynamic Correlation Analysis:** Identify relationships among system parameters and study patterns preceding device failures.
- 3. Model Development:** Implement and compare algorithms such as Isolation Forest, Autoencoders, and One-Class SVM for anomaly detection.
- 4. Performance Evaluation:** Assess models using metrics like Precision, Recall, and F1-score against a baseline system.
- 5. Insight Generation:** Derive explainable results by identifying the most influential parameters linked to failures.

By fulfilling these objectives, the project aims to build a scalable, reliable, and interpretable system that enhances predictive maintenance and device reliability.

LITERATURE REVIEW

Anomaly detection in device and system logs has become vital for ensuring reliability, minimizing downtime, and enabling predictive maintenance. Over time, research in this field has evolved from rule-based monitoring to intelligent, data-driven analysis.

1. Traditional and Statistical Approaches:

Early methods relied on statistical and threshold-based techniques, where alerts were triggered when certain parameters (e.g., CPU temperature or voltage) exceeded fixed limits. Although simple and interpretable, these systems often missed subtle deviations and required frequent manual updates. They lacked adaptability to dynamic environments and were not suitable for large-scale, heterogeneous device data.

2. Machine Learning-Based Methods:

The rise of machine learning enabled systems to learn normal behaviour from data.

- Isolation Forest (Liu et al., 2008) became popular for efficiently detecting anomalies in high-dimensional datasets without labelled data.
- One-Class SVM (Schölkopf et al., 2001) defined a boundary for normal data and classified deviations as anomalies.

These models improved automation and accuracy compared to rule-based methods but struggled with unstructured and temporal log data.

3. Deep Learning for Log Analysis:

Deep learning models introduced the ability to capture sequential and contextual patterns in logs.

- DeepLog (Du et al., 2017) used LSTM networks to model event sequences and detect abnormal patterns in real time.
- LogAnomaly (Meng et al., 2019) enhanced this by adding attention mechanisms to better capture dependencies among log entries.
- Autoencoders have also been used for unsupervised detection by reconstructing input data and flagging instances with high reconstruction errors, making them effective for identifying complex, non-linear anomalies.

4. Industry Applications and Trends:

Companies like Intel, IBM, and Dell have implemented predictive maintenance systems using anomaly detection to forecast hardware issues and reduce downtime. Intel's studies reported up to a 25% improvement in fault prediction, emphasizing the practical impact of such methods.

5. Research Gaps and Motivation for This Work:

Existing models face challenges such as:

- Handling unstructured, heterogeneous log formats,
- Adapting to dynamic system behaviour, and
- Providing explainable predictions for better interpretability.

This project builds upon works like DeepLog, LogAnomaly, and Autoencoder-based models to create an adaptable, interpretable, and scalable framework for device failure prediction through dynamic log correlation.

PROPOSED METHODOLOGY

The proposed methodology aims to develop a machine learning-based framework for detecting abnormal device behaviours and predicting failures through the correlation of system and failure logs. The process is organized into six key phases:

1. Data Understanding and Exploration

This phase focuses on analysing device log datasets that include attributes like timestamps, event types, error codes, and performance metrics. Exploratory Data Analysis (EDA) will be used to visualize trends, correlations, and failure patterns, helping identify the most relevant features for model development.

2. Data Preprocessing and Feature Engineering

Device logs are often noisy and incomplete, requiring thorough preprocessing. Key steps include data cleaning (handling missing or duplicate entries), normalization, and encoding of categorical variables.

Feature engineering will involve deriving metrics such as event frequency, rolling averages, and time-window representations to capture temporal patterns in device performance.

3. Correlation and Pattern Analysis

Statistical and ML-based correlation techniques (e.g., Pearson/Spearman coefficients) will be applied to uncover relationships between system parameters and failure events. Visualization tools like heatmaps and dependency graphs will help interpret these patterns and guide model input selection.

4. Model Development

Multiple algorithms will be implemented and compared for anomaly detection:

- **Isolation Forest**- Efficient for high-dimensional mixed data.
- **Autoencoders**- Neural networks that detect anomalies via reconstruction error.
- **One-Class SVM**- Learns the boundary of normal behaviour with limited failure data.

Models will be trained on normal logs and tested on datasets containing both normal and anomalous samples.

5. Model Evaluation and Validation

Models will be evaluated using metrics like Precision, Recall, F1-score, and ROC-AUC. A threshold-based baseline system will serve as a benchmark, and cross-validation will ensure consistent, reliable performance across datasets.

6. Interpretability and Insight Generation

To enhance transparency, SHAP analysis and feature importance methods will identify key parameters influencing anomaly detection. The final outputs will include:

- The trained anomaly detection model
- Correlation maps and visual insights
- A summary report highlighting predictive failure patterns and insights for preventive maintenance

In summary, this methodology integrates statistical analysis, machine learning, and interpretability tools to create a robust, explainable anomaly detection framework that improves device reliability and supports predictive maintenance.

Expected Outcomes

The proposed project aims to develop a data-driven, interpretable, and efficient anomaly detection framework capable of identifying early signs of device failures. By dynamically correlating system and failure logs, it seeks to transition from reactive fault detection to predictive maintenance.

1. An Anomaly Detection Model:

- A trained machine learning model that accurately detects deviations from normal device behaviour.
- Improved detection accuracy and reduced false alarms compared to traditional threshold-based monitoring.

2. Dynamic Correlation Insights:

- Identification of hidden relationships between log parameters and failure events.
- Generation of key failure indicators that assist engineers in proactive maintenance.

3. Performance Evaluation and Benchmarking:

- Model assessment using metrics like Precision, Recall, F1-score, and ROC-AUC.
- Benchmarking against conventional rule-based systems to validate performance gains.

4. Explainability and Interpretability:

- Use of explainable AI methods (e.g., SHAP or feature importance) to clarify anomaly causes.
- Enhanced model transparency and trust for real-world maintenance applications.

Overall, the project is expected to contribute to the development of a robust, interpretable, and scalable anomaly detection system that enhances device reliability, reduces downtime, and supports the vision of intelligent self-monitoring systems.

CHALLENGES AND LIMITATIONS

- 1. Data Quality and Availability:** Device logs are often noisy, incomplete, or inconsistent, with missing timestamps and redundant entries. Moreover, failure data is limited since real-world faults are rare, leading to class imbalance and reduced model accuracy.
- 2. High Dimensionality and Complexity:** Logs can include numerous parameters and unstructured text, making feature extraction and correlation analysis computationally intensive. Selecting relevant features without losing important information remains difficult.
- 3. Temporal Dependency Handling:** Failures may result from gradual changes rather than sudden anomalies. Capturing these temporal dependencies accurately requires sequence modelling, which increases model complexity.
- 4. Model Interpretability:** Advanced models like autoencoders or neural networks act as “black boxes”. Ensuring interpretability through methods such as SHAP or feature importance adds extra computation and design effort.
- 5. Generalization and Real-Time Constraints:** Models trained on one device type may not generalize across different hardware or environments. Additionally, deploying the system for real-time monitoring requires balancing speed, memory efficiency, and false alarm control.

In summary, while the proposed approach promises significant improvements in predictive maintenance, challenges such as limited labelled data, temporal complexity, and real-time deployment must be carefully addressed to ensure scalability, interpretability, and long-term effectiveness.

FUTURE SCOPE

The current project lays the foundation for a robust and interpretable device anomaly detection framework, but several directions can further enhance its capability and applicability in real-world environments.

- 1. Real-Time Monitoring:** Extend the model for live anomaly detection and instant alerts to enable proactive system maintenance and minimize downtime.
- 2. Broader Device Coverage:** Adapt the framework for varied device types, including servers, laptops, and IoT systems, to improve generalizability.
- 3. Advanced Temporal Modelling:** Explore deep learning architectures like LSTMs or Transformers to better capture evolving system patterns and long-term dependencies.
- 4. Improved Explainability:** Develop intuitive dashboards and visualization tools to make model decisions and correlations easier to interpret.
- 5. Continuous and Adaptive Learning:** Implement online learning to allow the system to update automatically with new data and changing device conditions.

In summary, the framework can evolve into a real-time, adaptive, and fully automated predictive maintenance system, contributing toward the vision of intelligent, self-healing devices.

REFERENCES

1. Liu, F. T., Ting, K. M., & Zhou, Z. H. (2008). *Isolation Forest*. Proceedings of the 8th IEEE International Conference on Data Mining (ICDM), 413–422.
2. Schölkopf, B., Platt, J., Shawe-Taylor, J., Smola, A. J., & Williamson, R. C. (2001). *Estimating the Support of a High-Dimensional Distribution*. Neural Computation, 13(7), 1443–1471.
3. Du, M., Li, F., Zheng, G., & Srikumar, V. (2017). *DeepLog: Anomaly Detection and Diagnosis from System Logs through Deep Learning*. Proceedings of the ACM Conference on Computer and Communications Security (CCS), 1285–1298.
4. Meng, W., Liu, Y., Zhang, S., Zhang, L., & Xu, M. (2019). *LogAnomaly: Unsupervised Detection of Sequential and Quantitative Anomalies in Unstructured Logs*. Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI), 4739–4745.
5. Chandola, V., Banerjee, A., & Kumar, V. (2009). *Anomaly Detection: A Survey*. ACM Computing Surveys (CSUR), 41(3), 1–58.
6. Ahmed, M., Mahmood, A. N., & Hu, J. (2016). *A Survey of Network Anomaly Detection Techniques*. Journal of Network and Computer Applications, 60, 19–31.
7. Intel Corporation. (2020). *Predictive Maintenance Using Machine Learning*. Intel IT Technical White Paper.
8. Dell Technologies. (2021). *AI-Driven Predictive Maintenance for IT Infrastructure*. Dell Technologies Report.